

MVLU COLLEGE

AI PRACTICAL NO. 9

AIM: Association Rule Mining

- Implement the Association Rule Mining algorithm (e.g., Apriori) to find frequent itemsets.
- Generate association rules from the frequent itemsets and calculate their support and confidence.
- Interpret and analyze the discovered association rules.

#CODE:

PRACTICAL 9: ASSOCIATION RULE MINING USING APRIORI

```
import pandas as pd
import matplotlib.pyplot as plt
from mlxtend.frequent_patterns import apriori, association_rules
```

1. LOAD DATASET

```
df = pd.read_csv("C:/Users/DELL/Desktop/siya/DATASET/new_test.csv")
```

```
print("==== DATASET ====")
print(df.head())
```

```
print("\nDataset Shape:", df.shape)
print("\nColumns:", df.columns.tolist())
```

2. PREPARE TRANSACTION DATA

```
data = df.copy()
```

```
# Convert age into categories
data["age_group"] = pd.cut(
    data["age"],
    bins=[0, 30, 40, 50, 60, 100],
    labels=[
        "Age <= 30",
        "Age 31-40",
        "Age 41-50",
        "Age 51-60",
        "Age > 60"
    ]
)
```

```
# Convert duration into categories
data["duration_group"] = pd.cut(
```

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```
data["duration"],
bins=4,
labels=[
    "Duration Low",
    "Duration Medium",
    "Duration High",
    "Duration Very High"
]
)

# Remove original numerical columns
data = data.drop(columns=["age", "duration"])

# Convert data into transaction format
basket = pd.get_dummies(data.astype(str)).astype(bool)

print("\n===== TRANSACTION DATA =====")
print(basket.head())

# 3. FIND FREQUENT ITEMSETS USING APRIORI

frequent_itemsets = apriori(
    basket,
    min_support=0.05,
    use_colnames=True
)

frequent_itemsets = frequent_itemsets.sort_values(
    "support",
    ascending=False
)

print("\n===== FREQUENT ITEMSETS =====")

print(
    frequent_itemsets[
        ["support", "itemsets"]
    ].head(20)
)

print("\nTotal Frequent Itemsets:",
      len(frequent_itemsets))
```

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4. GENERATE ASSOCIATION RULES

```
rules = association_rules(
    frequent_itemsets,
    metric="confidence",
    min_threshold=0.50
)

rules = rules.sort_values(
    "confidence",
    ascending=False
)

print("\n===== ASSOCIATION RULES =====")

print(
    rules[
        [
            "antecedents",
            "consequents",
            "support",
            "confidence",
            "lift"
        ]
    ].head(20)
)
```

5. DISPLAY TOP RULES

```
print("\n===== TOP ASSOCIATION RULES =====")

for _, row in rules.head(10).iterrows():

    antecedent = ", ".join(row["antecedents"])
    consequent = ", ".join(row["consequents"])

    print("\n", antecedent, "---->", consequent)

    print("Support:",
          round(row["support"], 3))

    print("Confidence:",
```

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```
round(row["confidence"], 3))
```

```
print("Lift:",  
      round(row["lift"], 3))
```

6. INTERPRET STRONGEST RULE

```
if len(rules) > 0:
```

```
    best = rules.iloc[0]
```

```
    print("\n===== INTERPRETATION =====")
```

```
    print(  
        "Strongest Rule:",  
        ", ".join(best["antecedents"]),  
        "---->",  
        ", ".join(best["consequents"])  
    )
```

```
    print("Support:",  
          round(best["support"], 3))
```

```
    print("Confidence:",  
          round(best["confidence"], 3))
```

```
    print("Lift:",  
          round(best["lift"], 3))
```

```
    if best["lift"] > 1:  
        print("Analysis: Positive association between the items.")
```

```
    elif best["lift"] == 1:  
        print("Analysis: The items are independent.")
```

```
    else:  
        print("Analysis: Negative association between the items.")
```

```
else:  
    print("\nNo association rules found.")
```

7. GRAPH

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```
if len(rules) > 0:
```

```
    plt.scatter(  
        rules["support"],  
        rules["confidence"]  
    )
```

```
    plt.xlabel("Support")  
    plt.ylabel("Confidence")  
    plt.title("Association Rules: Support vs Confidence")  
    plt.grid()
```

```
    plt.show()
```

```
# FINAL RESULT
```

```
print("\n===== FINAL RESULT =====")
```

```
print(  
    "Frequent Itemsets Found:",  
    len(frequent_itemsets)  
)
```

```
print(  
    "Association Rules Found:",  
    len(rules)  
)
```

```
print("\nPractical Completed Successfully!")
```

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#OUTPUT:

```
*IDLE Shell 3.13.2*
File Edit Shell Debug Options Window Help
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb 4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/DELL/Desktop/siya/T105_AI_Prac9.py =====
===== DATASET =====
   age  job  marital  education  ...  day_of_week  duration  campaign  poutcome
0   32    4         0          6  ...           3         131          5          1
1   37   10         3          6  ...           3         100          1          1
2   55    5         0          5  ...           2         131          2          1
3   44    2         1          0  ...           3          48          2          1
4   28    0         2          3  ...           0         144          2          1

[5 rows x 13 columns]

Dataset Shape: (8238, 13)

Columns: ['age', 'job', 'marital', 'education', 'default', 'housing', 'loan', 'contact', 'month', 'day_of_w
week', 'duration', 'campaign', 'poutcome']

===== TRANSACTION DATA =====
   job_0  ...  duration_group_Duration  Very High
0  False  ...                               False
1  False  ...                               False
2  False  ...                               False
3  False  ...                               False
4   True  ...                               False

[5 rows x 65 columns]

===== FREQUENT ITEMSETS =====
support      itemsets
39  0.867565      (poutcome_1)
18  0.827749      (loan_0)
14  0.798859      (default_0)
352  0.718742      (poutcome_1, loan_0)
258  0.681597      (poutcome_1, default_0)
237  0.661447      (loan_0, default_0)
20  0.635591      (contact_0)
6   0.603909      (marital_1)
1169  0.564943      (poutcome_1, loan_0, default_0)
239  0.532654      (contact_0, default_0)
143  0.529983      (poutcome_1, marital_1)
333  0.525492      (contact_0, loan_0)
17  0.524642      (housing_2)
383  0.512382      (poutcome_1, contact_0)
124  0.502064      (loan_0, marital_1)
45  0.463219      (duration_group_Duration Low)
120  0.456543      (marital_1, default_0)
324  0.454965      (poutcome_1, housing_2)
16  0.451809      (housing_0)
805  0.440884      (poutcome_1, loan_0, marital_1)

Total Frequent Itemsets: 3609

===== ASSOCIATION RULES =====
antecedents  ...  lift
7964  (month_4, duration_group_Duration Low, contact_1)  ...  1.147888
5688  (month_4, contact_1, marital_1)  ...  1.146596
2037  (month_4, contact_1)  ...  1.146234
```

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```
*IDLE Shell 3.13.2*
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===== ASSOCIATION RULES =====
antecedents    ...    lift
7964 (month_4, duration_group_Duration Low, contact_1) ... 1.147888
5688 (month_4, contact_1, marital_1) ... 1.146596
2037 (month_4, contact_1) ... 1.146234
6848 (duration_group_Duration Low, month_3, loan_0) ... 1.146014
8096 (housing_0, month_4, contact_1) ... 1.145477
8778 (month_4, contact_1, loan_0, marital_1) ... 1.145247
8853 (month_6, contact_1, age_group_Age 41-50) ... 1.145231
3263 (month_4, contact_1, loan_0) ... 1.144767
10249 (contact_0, duration_group_Duration Low, month_3, loan_0) ... 1.144628
4060 (month_4, contact_1, default_0) ... 1.143959
4563 (month_3, loan_0, marital_1) ... 1.143474
6348 (contact_0, month_3, loan_0, marital_1) ... 1.141998
6407 (month_4, contact_1, loan_0, default_0) ... 1.141959
10793 (month_3, campaign_1, loan_0) ... 1.141674
7931 (loan_0, month_6, contact_1, housing_0, marital_1) ... 1.140842
8035 (contact_1, default_1, loan_0, marital_1) ... 1.140793
4342 (duration_group_Duration Low, month_3) ... 1.140123
10132 (default_0, month_3, loan_0, marital_1) ... 1.139403
6968 (month_6, contact_1, campaign_1, loan_0) ... 1.139326
7311 (month_6, contact_1, default_1) ... 1.139064

[20 rows x 5 columns]

===== TOP ASSOCIATION RULES =====

month_4, duration_group_Duration Low, contact_1 ---> poutcome_1
Support: 0.059
Confidence: 0.996

Ln: 137 Col: 0
```

```
*IDLE Shell 3.13.2*
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month_4, duration_group_Duration Low, contact_1 ---> poutcome_1
Support: 0.059
Confidence: 0.996
Lift: 1.148

month_4, contact_1, marital_1 ---> poutcome_1
Support: 0.069
Confidence: 0.995
Lift: 1.147

month_4, contact_1 ---> poutcome_1
Support: 0.108
Confidence: 0.994
Lift: 1.146

duration_group_Duration Low, month_3, loan_0 ---> poutcome_1
Support: 0.063
Confidence: 0.994
Lift: 1.146

housing_0, month_4, contact_1 ---> poutcome_1
Support: 0.058
Confidence: 0.994
Lift: 1.145

month_4, contact_1, loan_0, marital_1 ---> poutcome_1
Support: 0.056
Confidence: 0.994
Lift: 1.145

Ln: 137 Col: 0
```

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```
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Confidence: 0.994
Lift: 1.145

month_6, contact_1, age_group_Age 41-50 ---> poutcome_1
Support: 0.056
Confidence: 0.994
Lift: 1.145

month_4, contact_1, loan_0 ---> poutcome_1
Support: 0.088
Confidence: 0.993
Lift: 1.145

contact_0, duration_group_Duration Low, month_3, loan_0 ---> poutcome_1
Support: 0.052
Confidence: 0.993
Lift: 1.145

month_4, contact_1, default_0 ---> poutcome_1
Support: 0.08
Confidence: 0.992
Lift: 1.144

===== INTERPRETATION =====
Strongest Rule: month_4, duration_group_Duration Low, contact_1 ---> poutcome_1
Support: 0.059
Confidence: 0.996
Lift: 1.148
Analysis: Positive association between the items.
```

